

Anshul Kolli

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EDUCATION

University of California, San Diego

B.S. in Computer Science; GPA: 3.95

Expected 2027

San Diego, CA

EXPERIENCE

Full Stack Developer Intern

Jun 2025 – Present

I Get Happy

Remote

- Engineered scalable backend features for a cross-platform application utilizing Firebase and GCP; integrated Twilio for automated crisis support routing.
- Built and deployed an AI chatbot leveraging OpenAI/Anthropic APIs and a custom RAG architecture for efficient, context-aware retrieval.

Co-founder

May 2024 – Present

SlugDining

Santa Cruz, CA

- Architected backend infrastructure and managed the primary data pipeline, supporting high-traffic user concurrency.
- Led a cross-functional engineering and design team to optimize UX and streamline Firebase database integrations.

Officer

May 2024 – June 2025

Santa Cruz AI Club

Santa Cruz, CA

- Organized UCSC's largest AI hackathon (200+ participants); developed workshops on GPU computing and deep learning

PROJECTS

Amazon Product Category Classifier | *PyTorch, Hugging Face, Scikit-learn, CUDA, NumPy, Pandas*

- Built end-to-end ML pipeline predicting product categories from 20K Amazon reviews using hybrid text-metadata features (TF-IDF + numeric) with sparse matrix optimization via SciPy
- Implemented and benchmarked multiple models: Multinomial Naive Bayes, Logistic Regression (L1/L2/ElasticNet), Linear SVM (grid search over C, loss functions), and fine-tuned DistilBERT
- Engineered PyTorch training loop with CUDA acceleration, custom cross-entropy/hinge losses, AdamW optimizer, and mixed-precision (fp16) for transformer fine-tuning

Memory-as-a-Service (MaaS) | *FastAPI, PostgreSQL, pgvector, Redis, Celery, Docker, sentence-transformers*

- Engineered a production-ready memory API prioritizing privacy-first PII redaction and a purpose-binding policy engine (consent, TTL, audit logs).
- Architected backend to handle 1000+ RPS reads (P95 120ms) and 200+ RPS writes using an async architecture, caching, and connection pooling.
- Implemented hybrid search combining pgvector cosine similarity and PostgreSQL full-text search via Reciprocal Rank Fusion (RRF).

GeminiRAG | *Python, CUDA, PyTorch, FAISS, Hugging Face*

- Engineered high-performance RAG with GPU-accelerated embeddings; 145× throughput (4 → 580 chunks/s) via CUDA parallelism
- Reduced VRAM usage 75% using quantization while boosting throughput 40%

SlugDining App | *Flutter, Dart, Firebase, Web Scraping*

- Architected a highly scalable backend to serve 3000+ concurrent users.
- Optimized database queries and implemented menu caching, reducing overall application load times by 40%.

People Tracker | *Next.js, TypeScript, Firebase, Grok AI*

- Built an AI-powered CRM with serverless Cloud Functions for real-time message analysis and sentiment detection.
- Implemented an AI vision model to automatically extract contact data from screenshots across multiple messaging platforms.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, CUDA, JavaScript/TypeScript, Dart

AI/ML: PyTorch, Hugging Face Transformers, FAISS, Scikit-learn, NumPy, Pandas, sentence-transformers

Infrastructure: Docker, Kubernetes, GCP, Azure, Firebase, PostgreSQL, Redis, FastAPI, asyncio

Tools: Git, Matplotlib, Seaborn, React.js, Next.js, Flutter